

Proof. Let $x \in \mathbb{R}$ satisfy $x \geq -1$ and let $n \in \mathbb{Z}_{\geq 0}$.

1. Preliminary observations. Since $x \geq -1$ we have $1 + x \geq 0$, so $(1 + x)^n$ is well defined for every non-negative integer n .

- $n = 0$: $(1 + x)^0 = 1 = 1 + 0 \cdot x$; equality holds.
- $n = 1$: $(1 + x)^1 = 1 + x = 1 + 1 \cdot x$; equality holds.

2. Base of induction ($n = 2$).

$$(1 + x)^2 = 1 + 2x + x^2 \geq 1 + 2x,$$

because $x^2 \geq 0$. Hence the inequality holds for $n = 2$.

3. Inductive hypothesis. Assume that for some integer $k \geq 2$ we have, for every $x \geq -1$,

$$(1 + x)^k \geq 1 + kx. \quad (\text{IH})$$

4. Inductive step ($k \rightarrow k + 1$). Multiplying (IH) by the non-negative factor $1 + x$ preserves the inequality:

$$(1 + x)^{k+1} = (1 + x)^k(1 + x) \quad (1)$$

$$\geq (1 + kx)(1 + x). \quad (1)$$

Expanding the right-hand side gives

$$(1 + kx)(1 + x) = 1 + (k + 1)x + kx^2. \quad (2)$$

Since $k \geq 2$ we have $k \geq 0$ and $x^2 \geq 0$, whence

$$kx^2 \geq 0. \quad (3)$$

Adding the non-negative term kx^2 to $1 + (k + 1)x$ cannot decrease the value, so from (2) and (3) we obtain

$$(1 + x)^{k+1} \geq 1 + (k + 1)x. \quad (4)$$

Thus the inequality holds for the exponent $k + 1$.

5. Conclusion by induction. The base case $n = 2$ is true, and the inductive step shows that truth for $n = k$ ($k \geq 2$) implies truth for $n = k + 1$. Together with the already verified cases $n = 0$ and $n = 1$, we have

$$(1 + x)^n \geq 1 + nx \quad \text{for all } n \in \mathbb{Z}_{\geq 0}, x \geq -1.$$

6. Equality condition. In the inductive step the only potentially strict term is the non-negative quantity kx^2 appearing in (2). Equality in (4) therefore requires

$$kx^2 = 0.$$

For $k \geq 2$ this forces $x = 0$. The previously treated exponents give equality for any x when $n = 0$ or $n = 1$. Consequently,

$$(1 + x)^n = 1 + nx \iff n = 0 \text{ or } n = 1 \text{ or } x = 0.$$

Hence the theorem is proved. $\square$

$\square$